B **03000CS304052002** **Pages: 3**

JYOTHI ENGG. COLLEGE LIBRARY
THRISSUR EDUCATIONAL TRUST
LEAD KINDLY LIGHT
CHERUTHURUTHY

Reg No.:_____ Name:_____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Sixth Semester B.Tech Degree Regular and Supplementary Examination July 2021

Course Code: CS304
Course Name: COMPILER DESIGN

Max. Marks: 100 Duration: 3 Hours

PART A

Answer all questions, each carries 3 marks. Marks

1 Write a regular expression to denote the language of all strings of a's and b's with an even number of a's followed by an odd number of b's. (3)

2 Distinguish between front end and back end of a compiler. (3)

3 Show that the following grammar is ambiguous : (3)

$S \rightarrow i\,C\,t\,S$

$S \rightarrow i\,C\,t\,S\,e\,S$

$S \rightarrow a$

$C \rightarrow b$

4 Explain backtracking with an example. (3)

PART B

Answer any two full questions, each carries 9 marks.

5 a) Eliminate left recursion from the following grammar: (4)

$S \rightarrow Aa \mid b$

$A \rightarrow Ac \mid Sd \mid h$

b) Construct a recursive descent parser for the following grammar (5)

$E \rightarrow TE'$

$E' \rightarrow +TE' \mid \varepsilon$

$T \rightarrow FT'$

$T' \rightarrow *FT' \mid \varepsilon$

$F \rightarrow (E) \mid id$

6 Explain in detail the various phases of a compiler with a neat diagram. Illustrate (9)

the output of each phase for the input $x = 2 * a + b$, where a and b are float variables.

7 a) Check if following grammar is LL(1) by constructing a parse table: (5)

$S \rightarrow (L) \mid a$

$L \rightarrow SL'$

$L' \rightarrow , SL' \mid \varepsilon$

Note that ',' is a terminal and ε is the empty string.

b) Explain bootstrapping. (4)

PART C

Answer all questions, each carries 3 marks.

8 Determine the FIRST and FOLLOW sets for the non-terminals in the following grammar: (3)

$S \rightarrow AA$

$A \rightarrow aA \mid b$

9 Define an operator grammar. Give an example (3)

10 Distinguish between S-attributed definitions and L-attributed definitions. (3)

11 What is type checking? What are its two types? (3)

PART D

Answer any two full questions, each carries 9 marks.

12 Construct the SLR(1) parsing table for the following grammar: (9)

$E \rightarrow T + E \mid T$

$T \rightarrow id$

13 a) Write a syntax directed translation scheme for a simple desk calculator. (3)

b) What sequence of moves are made by a bottom up parser on the input 23*5+4 using the translations in part (a)? (6)

14 a) Consider the following grammar: (6)

$S \rightarrow a \mid (T)$

$T \rightarrow T, S \mid S$

For the string *(a, (a, a))*, indicate how a parse tree is constructed by a shift reduce parser using a rightmost derivation.

b) Explain the bottom- up evaluation of S- attributed definitions. (3)

PART E

Answer any four full questions, each carries 10 marks.

15 a) What is heap allocation strategy? (3)

b) What is an activation record? Explain its structure with a figure. (7)

16 Write quadruples, triples and indirect tuples for the expression (10)

(a + b) * (b + c) + (a + b + c)

17 a) Write a syntax directed translation scheme that generates three address code for Boolean expressions. (7)

b) Distinguish between static and dynamic storage allocation. (3)

18 a) Write the algorithm for identifying the basic blocks from a sequence of three address code statements. (5)

b) Construct the DAG for the following basic block (5)

D := B * C

E := A + B

B := B * C

A := E – D

19 For the following C statement, write the three-address code. (10)

X := A – B + C – D + E – F

Convert the three-address code into machine code.

20 a) How do algebraic laws help in optimizing basic blocks? (4)

b) Write the code generation algorithm. (6)

****